



11867 - JWST Imaging of GRB Host Galaxies at $z=6-9$

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		NIRCam Imaging	(1) GRB050904
	2		NIRCam Imaging	(2) GRB080913
	3		NIRCam Imaging	(3) GRB090423
	4		NIRCam Imaging	(4) GRB090429B
	5		NIRCam Imaging	(5) GRB100905A
	6		NIRCam Imaging	(6) GRB120521C
	7		NIRCam Imaging	(7) GRB120923A
	8		NIRCam Imaging	(8) GRB140515A
	9		NIRCam Imaging	(9) GRB210905A
	10		NIRCam Imaging	(10) GRB240218A

ABSTRACT

Long Gamma-ray bursts (GRBs) are powerful probes of the early universe thanks to their remarkable brightness. Many GRBs have been spectroscopically and photometrically identified at high redshifts, up to $z \sim 9.4$, through observations of their afterglows with optical and near-infrared ground-based telescopes. Since GRBs originate from the deaths of massive stars whose lifetime is short, the evolution of the GRB formation rate should generally follow the cosmic star formation rate (SFR) density measured from deep galaxy surveys. However, at high redshifts, GRB-based SFR densities show a significant discrepancy when compared to those measured from observed galaxies in these surveys. The physical origin of this discrepancy remains unclear, since the properties of GRB host galaxies at $z > 6$ are poorly understood, leaving this issue unresolved for over a decade.

In this proposal, we aim to conduct deep NIRCam imaging observations targeting 10 GRBs at $z = 6-9$. Our goal is to detect the UV continuum of the host galaxies or, in the absence of a detection, establish stringent upper limits down to $M_{UV} = -16.4$ mag (3σ) in the most sensitive NIRCam band, F150W2. Additionally, observations with the longer-wavelength F277W/F356W/F444W bands will allow us to investigate the degree of extinction and dust-obscured star formation. Even non-detections would suggest that GRBs at $z > 6$ (and the bulk of star formation) occur in extremely faint galaxies that are missed in the deep galaxy surveys. Based on these results, we will characterize the SEDs of the host galaxies of GRBs at $z = 6-9$ and discuss the origin of the tension between the GRB-based and galaxy-based cosmic SFR densities.

OBSERVING DESCRIPTION

We will conduct NIRCam F150W2 and F277W/F356W/F444W imaging observations targeting 10 GRBs at $z = 6-9$, to characterize their host galaxies. The exposure time is set to 1 hour, reaching the 5σ limiting magnitudes of 30.0 mag in F150W2 and 28.1 mag in F277W/F356W/F444W for a point source, sufficient to achieve our science goal.

Proposal 11867 - Targets - JWST Imaging of GRB Host Galaxies at z=6-9

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	GRB050904	RA: 00 54 50.8800 (13.7120000d) Dec: +14 05 9.90 (14.08608d) Equinox: J2000 Comments: Category=Unidentified Description=[X-ray sources]		
	(2)	GRB080913	RA: 04 22 54.7200 (65.7280000d) Dec: -25 07 46.00 (-25.12944d) Equinox: J2000 Comments: Category=Unidentified Description=[X-ray sources]		
	(3)	GRB090423	RA: 09 55 33.0800 (148.8878333d) Dec: +18 08 58.90 (18.14969d) Equinox: J2000 Comments: Category=Unidentified Description=[X-ray sources]		
	(4)	GRB090429B	RA: 14 02 40.0500 (210.6668750d) Dec: +32 10 14.32 (32.17064d) Equinox: J2000 Comments: Category=Unidentified Description=[X-ray sources]		
	(5)	GRB100905A	RA: 02 06 12.0400 (31.5501667d) Dec: +14 55 45.80 (14.92939d) Equinox: J2000 Comments: Category=Unidentified Description=[X-ray sources]		
	(6)	GRB120521C	RA: 14 17 8.8000 (214.2866667d) Dec: +42 08 41.50 (42.14486d) Equinox: J2000 Comments: Category=Unidentified Description=[X-ray sources]		
	(7)	GRB120923A	RA: 20 15 11.1100 (303.7962917d) Dec: +06 13 13.40 (6.22039d) Equinox: J2000 Comments: Category=Unidentified Description=[X-ray sources]		
	(8)	GRB140515A	RA: 12 24 15.5100 (186.0646250d) Dec: +15 06 16.80 (15.10467d) Equinox: J2000 Comments: Category=Unidentified Description=[X-ray sources]		

Proposal 11867 - Targets - JWST Imaging of GRB Host Galaxies at z=6-9

	(9)	GRB210905A	RA: 20 36 11.6400 (309.0485000d) Dec: -44 26 24.00 (-44.44000d) Equinox: J2000
	Comments: Category=Unidentified Description=[X-ray sources]		
	(10)	GRB240218A	RA: 10 47 11.2320 (161.7968000d) Dec: +01 16 34.68 (1.27630d) Equinox: J2000
	Comments: Category=Unidentified Description=[X-ray sources]		

Proposal 11867 - Observation 1 - JWST Imaging of GRB Host Galaxies at z=6-9

Observation	Proposal 11867, Observation 1										Fri Mar 13 22:03:43 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRCam Imaging											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 1:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(1)	GRB050904	RA: 00 54 50.8800 (13.7120000d) Dec: +14 05 9.90 (14.08608d) Equinox: J2000									
	Comments: Category=Unidentified Description=[X-ray sources]											
Template	Module		Subarray				Target Placement					
	ALL		FULL				Module B center (small extended source)					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions		
	1	INTRAMODULEBOX		4		STANDARD				1		
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID		
	1	F150W2	F277W	SHALLOW2	5	1	4	4	944.836			
	2	F150W2	F356W	SHALLOW2	5	1	4	4	944.836			
	3	F150W2	F444W	SHALLOW2	5	2	8	4	1932.619			
Special Requirements	Background Limited. Background no more than 40th percentile above minimum											
	Fiducial Point Override NRCBS_FULL											

Proposal 11867 - Observation 2 - JWST Imaging of GRB Host Galaxies at z=6-9

Observation	Proposal 11867, Observation 2										Fri Mar 13 22:03:43 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	GRB080913	RA: 04 22 54.7200 (65.7280000d) Dec: -25 07 46.00 (-25.12944d) Equinox: J2000								
	Comments: Category=Unidentified Description=[X-ray sources]										
Template	Module		Subarray				Target Placement				
	ALL		FULL				Module B center (small extended source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F150W2	F277W	SHALLOW2	5	1	4	4	944.836		
	2	F150W2	F356W	SHALLOW2	5	1	4	4	944.836		
	3	F150W2	F444W	SHALLOW2	5	2	8	4	1932.619		
Special Requirements	Background Limited. Background no more than 40th percentile above minimum Fiducial Point Override NRCBS_FULL										

Proposal 11867 - Observation 3 - JWST Imaging of GRB Host Galaxies at z=6-9

Observation	Proposal 11867, Observation 3										Fri Mar 13 22:03:43 GMT 2026
	Diagnostic Status: Warning										
Diagnostics	Observing Template: NIRCам Imaging										
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(3)	GRB090423	RA: 09 55 33.0800 (148.8878333d) Dec: +18 08 58.90 (18.14969d) Equinox: J2000								
Template	Comments: Category=Unidentified Description=[X-ray sources]										
	Module		Subarray				Target Placement				
Dithers	ALL		FULL				Module B center (small extended source)				
	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
Spectral Elements	1	INTRAMODULEBOX		4		STANDARD				1	
	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F150W2	F277W	SHALLOW2	5	1	4	4	944.836		
	2	F150W2	F356W	SHALLOW2	5	1	4	4	944.836		
Special Requirements	3	F150W2	F444W	SHALLOW2	5	2	8	4	1932.619		
	Background Limited. Background no more than 40th percentile above minimum Fiducial Point Override NRCBS_FULL										

Proposal 11867 - Observation 4 - JWST Imaging of GRB Host Galaxies at z=6-9

Observation	Proposal 11867, Observation 4										Fri Mar 13 22:03:43 GMT 2026
	Diagnostic Status: Warning										
Diagnostics	Observing Template: NIRCam Imaging										
	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(4)	GRB090429B	RA: 14 02 40.0500 (210.6668750d) Dec: +32 10 14.32 (32.17064d) Equinox: J2000								
Template	Comments: Category=Unidentified Description=[X-ray sources]										
	Module		Subarray				Target Placement				
Dithers	ALL		FULL				Module B center (small extended source)				
	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
Spectral Elements	1	INTRAMODULEBOX		4		STANDARD				1	
	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F150W2	F277W	SHALLOW2	5	1	4	4	944.836		
	2	F150W2	F356W	SHALLOW2	5	1	4	4	944.836		
Special Requirements	3	F150W2	F444W	SHALLOW2	5	2	8	4	1932.619		
	Background Limited. Background no more than 40th percentile above minimum Fiducial Point Override NRCBS_FULL										

Proposal 11867 - Observation 5 - JWST Imaging of GRB Host Galaxies at z=6-9

Observation	Proposal 11867, Observation 5										Fri Mar 13 22:03:43 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 5:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(5)	GRB100905A	RA: 02 06 12.0400 (31.5501667d) Dec: +14 55 45.80 (14.92939d) Equinox: J2000								
	Comments: Category=Unidentified Description=[X-ray sources]										
Template	Module		Subarray				Target Placement				
	ALL		FULL				Module B center (small extended source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F150W2	F277W	SHALLOW2	5	1	4	4	944.836		
	2	F150W2	F356W	SHALLOW2	5	1	4	4	944.836		
	3	F150W2	F444W	SHALLOW2	5	2	8	4	1932.619		
Special Requirements	Background Limited. Background no more than 40th percentile above minimum										
	Fiducial Point Override NRCBS_FULL										

Proposal 11867 - Observation 6 - JWST Imaging of GRB Host Galaxies at z=6-9

Observation	Proposal 11867, Observation 6										Fri Mar 13 22:03:43 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(6)	GRB120521C	RA: 14 17 8.8000 (214.2866667d) Dec: +42 08 41.50 (42.14486d) Equinox: J2000								
	Comments: Category=Unidentified Description=[X-ray sources]										
Template	Module			Subarray			Target Placement				
	ALL			FULL			Module B center (small extended source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F150W2	F277W	SHALLOW2	5	1	4	4	944.836		
	2	F150W2	F356W	SHALLOW2	5	1	4	4	944.836		
	3	F150W2	F444W	SHALLOW2	5	2	8	4	1932.619		
Special Requirements	Background Limited. Background no more than 40th percentile above minimum Fiducial Point Override NRCBS_FULL										

Proposal 11867 - Observation 7 - JWST Imaging of GRB Host Galaxies at z=6-9

Observation	Proposal 11867, Observation 7										Fri Mar 13 22:03:43 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(7)	GRB120923A	RA: 20 15 11.1100 (303.7962917d) Dec: +06 13 13.40 (6.22039d) Equinox: J2000								
	Comments: Category=Unidentified Description=[X-ray sources]										
Template	Module		Subarray				Target Placement				
	ALL		FULL				Module B center (small extended source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F150W2	F277W	SHALLOW2	5	1	4	4	944.836		
	2	F150W2	F356W	SHALLOW2	5	1	4	4	944.836		
	3	F150W2	F444W	SHALLOW2	5	2	8	4	1932.619		
Special Requirements	Background Limited. Background no more than 40th percentile above minimum Fiducial Point Override NRCBS_FULL										

Proposal 11867 - Observation 8 - JWST Imaging of GRB Host Galaxies at z=6-9

Observation	Proposal 11867, Observation 8										Fri Mar 13 22:03:43 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(8)	GRB140515A	RA: 12 24 15.5100 (186.0646250d) Dec: +15 06 16.80 (15.10467d) Equinox: J2000								
	Comments: Category=Unidentified Description=[X-ray sources]										
Template	Module		Subarray				Target Placement				
	ALL		FULL				Module B center (small extended source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F150W2	F277W	SHALLOW2	5	1	4	4	944.836		
	2	F150W2	F356W	SHALLOW2	5	1	4	4	944.836		
	3	F150W2	F444W	SHALLOW2	5	2	8	4	1932.619		
Special Requirements	Background Limited. Background no more than 40th percentile above minimum Fiducial Point Override NRCBS_FULL										

Proposal 11867 - Observation 9 - JWST Imaging of GRB Host Galaxies at z=6-9

Observation	Proposal 11867, Observation 9										Fri Mar 13 22:03:43 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(9)	GRB210905A	RA: 20 36 11.6400 (309.0485000d) Dec: -44 26 24.00 (-44.44000d) Equinox: J2000								
	Comments: Category=Unidentified Description=[X-ray sources]										
Template	Module				Subarray				Target Placement		
	ALL				FULL				Module B center (small extended source)		
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F150W2	F277W	SHALLOW2	5	1	4	4	944.836		
	2	F150W2	F356W	SHALLOW2	5	1	4	4	944.836		
	3	F150W2	F444W	SHALLOW2	5	2	8	4	1932.619		
Special Requirements	Background Limited. Background no more than 40th percentile above minimum Fiducial Point Override NRCBS_FULL										

Proposal 11867 - Observation 10 - JWST Imaging of GRB Host Galaxies at z=6-9

Observation	Proposal 11867, Observation 10										Fri Mar 13 22:03:43 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(10)	GRB240218A	RA: 10 47 11.2320 (161.7968000d) Dec: +01 16 34.68 (1.27630d) Equinox: J2000								
	Comments: Category=Unidentified Description=[X-ray sources]										
Template	Module		Subarray				Target Placement				
	ALL		FULL				Module B center (small extended source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F150W2	F277W	SHALLOW2	5	1	4	4	944.836		
	2	F150W2	F356W	SHALLOW2	5	1	4	4	944.836		
	3	F150W2	F444W	SHALLOW2	5	2	8	4	1932.619		
Special Requirements	Background Limited. Background no more than 40th percentile above minimum Fiducial Point Override NRCBS_FULL										